Solar PV Parameter Extraction Using GWO-PSO Algorithm

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2024 International Conference on Sustainable Power & Energy (ICSPE) | 979-8-3503-6605-1/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICSPE62629.2024.10924344

***Abstract*— In this paper, the proposed optimization approach is hybrid between the Grey Wolf Optimizer and the Particle Swarm Optimization, where it is utilized for extraction parameters in single-diode photovoltaic models that play an important role in accurately simulating and analyzing solar PV systems. These model parameters are highly nonlinear, posing a challenge in achieving the accurate estimation. Accurate values play an important role in the optimization of PV system performance and design. Hybrid GWO-PSO does the initial global search to identify promising regions of the solution space by using the exploration ability of GWO and refines those regions by utilizing the convergence strengths of PSO for the optimal values of the parameters. In order to enhance the precision further, the Nelder-Mead method is applied as a local search in fine-tuning the final solutions. Secondly, a penalty mechanism is there which deals with the negative values of R^2 scores and tries that the model should be more predictable and not get into solutions at low quality. For added robustness, we're using a Huber loss function in the objective reducing the effect of outliers towards the estimation of parameters involved. The proposed hybrid methodology is tested on real data of PV I-V for results evaluation using error metrics such as RMSE, MAE, and R^2. This will also reflect that the accuracy for the estimation of the algorithm is better than if applied GWO-PSO or PSO separately. Better parameter estimations lead to enhanced modeling of PV systems in multiple environmental conditions and push solar PV technology further both theoretically and practically. This, therefore, suggests an efficient, accurate extraction of PV parameters, which promises a better performance monitoring and system optimization within the solar energy industry.**

Keywords—component, formatting, style, styling, insert (*key words****)***

I. INTRODUCTION (*HEADING 1*)

The more people realize that a healthy environment requires preservation, the greater the demand for energy globally. Among the promising technologies related to clean energy generation is solar photovoltaic technology, which converts sunlight into electricity. Being a source of little environmental impact with such abundant availability, sunlight has lately become one of the key elements of new renewable energy policies. Efficiency and reliability are the main aspects that influence the economic viability and environmental benefits of the PV systems. This is the reason why research on these systems continues to be ongoing in the improvement of design, control, and performance[1][2].

Further, outliers in measured data, either due to sensor inaccuracy or environmental anomalies, can affect the extraction of PV parameters. The distortion of the parameter values will lead to an inaccurate representation of the I-V characteristics. Therefore, there is a need for accurate and outlier-resistant parameter extraction techniques that ensure the reliability of PV models under diverse conditions[3][4].

The heart of a PV system is the PV cell. The performance of a PV cell is generally described by the current-voltage characteristics. These describe the amount of electrical power produced under varying conditions such as changes in the intensity of sunlight, temperature, and shading. In an accurate modeling of the PV system, five key parameters of the single-diode model are to be determined: photocurrent Iph, saturation current I0, series resistance Rs, shunt resistance Rsh, and the ideality factor n. All these parameters capture the nonlinear behavior of PV cells and thus play a significant role in an accurate simulation and optimization process. However, this task is very complex due to the nonlinear nature of the PV system[5][6].

The accurate extraction of parameters for a PV system is crucial since it affects the prediction of the output for the system under real operating conditions. Traditional analytical methods used for parameter extraction present problems, especially with nonlinear PV systems. The iterative calculation-based methods with initial estimates often lead to inaccuracies and are extremely sensitive to the initial guess. Since the parameters in PV models vary due to the influences from changes in environmental conditions such as temperature and irradiance, adaptive robust methods become necessary[4].

These limitations make researchers interested to work on metaheuristic optimization techniques, which aptly suit nonlinear complex problems. Metaheuristics algorithms are inspired by models of natural or social process like animal behaviors or the strategy of evolution. Those algorithms generally avoid limitations due to initial guesses and smoothly traverse complex solution landscapes for efficient search in the space of solutions[7].

PSO and GWO are some of the widely known metaheuristic algorithms. They have been used successfully to solve many engineering problems and, therefore, the extraction of parameters in PV problems as well.

- Particle Swarm Optimization (PSO)

  PSO is based on the social behavior of birds or fish, where particles in a swarm move through the solution space, influenced by their personal best-known positions and the global best-known position. The advantages of PSO are simplicity, speed convergence, and not depending on derivative information. However, PSO sometimes converges prematurely in complex solution spaces and reaches suboptimal solutions[7][8][9].

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- Grey Wolf Optimizer (GWO)

  GWO is motivated by the hunting behavior of grey wolves, where there is a hierarchical structure wherein wolves (agents) cooperate to find prey (optimal solutions). GWO has strong exploration capabilities that help it avoid local optima during the search process. However, GWO can lack fine-grained exploitation, which limits its ability to refine solutions precisely near the global optimum[5][10].

Both of these algorithms have immense potential in the extraction of PV parameter estimation, yet their shortcomings may demand a hybridization that combines their strengths to achieve the best. This is the motivation for the hybrid GWO-PSO approach, where exploration and exploitation are carefully balanced to achieve higher accuracy in parameter estimation.

A. Hybrid GWO-PSO Methodology for PV Parameter Extraction

The hybrid GWO-PSO method is proposed to integrate the merits of GWO in terms of exploration, along with convergence abilities of PSO. In the proposed hybrid method, first, GWO explores the solution space profoundly, hence discovering promising regions in the search space without losing the optima. Hence, preliminary exploration reduces the possibility of premature convergence, which is one of the most prominent problems associated with PSO[11][12].

After the GWO selects some promising regions, in the next step with rapid convergence of PSO, this hybrid is successful in obtaining a level of precision concerning parameter estimation at further refinements within such search regions. This Nelder-Mead method aims for higher accuracy of a so far discovered solution that was proposed as the next implementation for performing the final phases of a local search after the entire optimization. It finds application mainly at fine-tuning of the solution in order to minimally correct small discrepancies of predicted versus actual I-V characteristics[13].

The hybrid approach uses mechanisms to adjust for data irregularities and outliers as well. Specifically, it penalizes if the R^2 score is negative so that the model's generated predictions are plausible. Further, it makes use of the Huber loss during optimization because it has lesser sensitivity to outliers than others; this further makes the model more robust for noisy data.

Here, a generic comparison of independent GWO, PSO, and the hybrid approach from GWO and PSO is presented to demonstrate that the hybrid method is fit for acquiring more accurate PV model parameters. This research is expected to contribute towards the body of knowledge in solar energy by providing a more reliable and efficient method to model PVs. Accurate models of PV are important in the design of systems, real-time monitoring, and optimization performance, as the hybrid approach between GWO-PSO can easily provide a practical solution for researchers and professionals in this field.

B. Objectives of this research

In this paper, the main aim is to design an accurate and robust method for extracting PV parameters using a hybrid GWO-PSO optimization algorithm.

- Improve the accuracy of PV parameter estimation that will allow better prediction of I-V characteristics under various operating conditions.
- Find a robust solution that is more reliable in the presence of outliers or noise in the data.
- Improving the convergence speed and stability of the optimization process through the combination of exploration strength from GWO and exploitation strength from PSO.

This introduction will help explain the relevance of PV parameter extraction, the extent to which the current methods stand today, and why it is employed in this hybrid approach of GWO and PSO. This paper balances both exploration and exploitation and further reinforces the general robustness by also leading into further information on methodology, results, and analysis in subsequent parts of the paper.

II. Modelling of Solar PV Parameter Extraction

The single-diode model (SDM) is widely accepted and practical due to its balance of accuracy and simplicity. It models the PV cell as an ideal current source in parallel with a diode, including additional resistances to account for real-world losses. The SDM has five key parameters—photo-generated current photo-generated current (I_{ph}), saturation current (I_0), series resistance (Rs), shunt resistance (R_{sh}), and the diode ideality factor (n). These parameters collectively capture the nonlinear behavior of the PV cell, including the effects of various environmental factors, such as temperature and irradiance[2].

The equation governing the I-V (current-voltage) characteristics of the PV cell in the single-diode model is given by:

$$I = I_{ph} - I_0\left(\exp\left(\frac{(V + IR_s)}{(nV_t)}\right) - 1\right) - \frac{(V + IR_s)}{R_{sh}} \quad (1)$$

- I is the output current of the PV cell
- V is the output voltage of the PV cell
- V_t is the thermal voltage of the cell, calculated as $V_t = kT/q$, with k as Boltzmann's constant, T as the cell temperature in Kelvin, and q as the electron charge.

Each of the parameters plays a critical role in defining the PV cell behavior:

- Photo-Generated Current (Iph)): It represents the current produced by the incident sunlight. It is directly proportional to the solar irradiance and varies with temperature.
- Saturation Current (Io): It is a reflection of the leakage current of the diode under reversed bias, which is temperature-dependent, thus affecting the cell's open-circuit voltage.
- Series Resistance (Rs): This accounts for the internal resistance of the PV cell material and connections, which would otherwise influence the slope of the I-V curve at higher current values.

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- Shunt Resistance (Rsh): Models the leakage current through the cell. Higher Rsh indicates that leakage is smaller, and consequently, it gives better performance at lower voltages.
- Diode Ideality Factor (n): This describes the recombination mechanism in the diode and controls how steep the I-V curve will become. The n values for most devices are found between 1 and 2.

Accurately determining these parameters is essential for effective PV system modeling and optimization. However, due to their interdependencies and nonlinear behavior, extracting these parameters is a challenging task.

$$SSE = \sum_{i=1}^{N} (I_{exp,i} - I_{calc,i}\ \ (Iph, I0, Rs, Rsh, n))^2 \quad (2)$$

- ($I_{exp,i}$) represents the experimentally measured current at the i_{th} data point,
- ($I_{calc,i}$) is the current calculated from the model at the same data point,
- (N) is the total number of data points.

The minimization of the objective function identifies parameters corresponding to the best fit of the model to the experimental data ensuring the latter accurately reflect the real world performance of the PV cell.

A. Challenges in PV Parameter Extraction

- Nonlinearity: The single-diode equation is nonlinear, and parameters can be extracted as an opti-mization problem.
- Interdependence of Parameters: Parameters such as Rs and Rsh affect each other, thus requiring complicated techniques to untangle their contributions accurately.
- Sensitivity to initial conditions.
- Traditional optimization algorithms always require initial uesses. If these guesses are far from the global optimum, they can get stuck in a local optimum.
- Environmental Variability: Parameters are sensitive to variations in both temperature and irradiance so require a strong method that can take into onsideration changes in external conditions. Outliers and measurement noise: There are typically outliers and noise present in experimental data that **slightly** skew the parameter extraction process.

B. Objective function with penalty

$$Objective\ Function = \sum_{i=j}^{N} (I_{exp,i}\ -\ I_{calc,i})^2 + Penalty \quad (3)$$

where the penalty is defined as:

$$Penalty = \begin{cases} 0, \ if\ R^2 \geq 0 \\ 1 * 10^6 * |R^2|, \ if R^2 < 0 \end{cases} \quad (4)$$

This approach limits the algorithm to not visiting certain regions of the solution space that provide unrepresentative models of the PV cell's behavior. Therefore, the extracted parameters will be more reliable and robust[5].

Where:

$I_{exp,i}$: Experimental current data

$I_{calc,i}$: Current calculated from the model

Thermal Voltage:

$$V_t = (k * T)/q \quad (5)$$

Where:

k: Boltzmann constant ($1.380649e^{-23}$ J/K)

T: Temperature in Kelvin

q: Elementary charge ($1.602176634e^{-19}$ C)

III. Optimization Process and Hybrid GWO-PSO Algorithm

This article introduces a Hybrid Grey Wolf Optimizer and Particle Swarm Optimization technique for the effective extraction of photovoltaic parameters. This hybrid methodology harnesses the advantages of both algorithms by attaining superior convergence rates and precision at obtaining optimal parameters.

The proposed hybridization algorithm, called Grey Wolf Optimizer (GWO) and Particle Swarm Optimization (PSO), is a powerful optimization framework. Such a hybrid approach was capable of effectively balancing the exploration capabilities of GWO with the exploitation strength of PSO. Generally speaking, such a hybrid approach would guarantee more balance between global and local search throughout the process of parameter optimization; thus, the technique suited such applications of solar PV parameter extraction.

A. GWO-PSO Hybrid Equation

The update mechanism for the hybrid algorithm is defined as follows[14]:

1. Grey Wolf Optimizer (GWO) Update Rule:

$$X(t+1) = (X_\alpha + X_\beta + X_\delta) / 3 \quad (6)$$

where:

$X(t+1)$ is the position of the current solution at iteration $(t+1)$.

$X_\alpha, X_\beta\ and\ X_\gamma$ represent the best, second best, and third best solutions at the current iteration, respectively.

Particle Swarm Optimization (PSO) Update Rule:

$$v(t+1) = w * v(t) + c_1 * r_1 * (P_best - X(t)) + c_2 * r_2 * (G_best - X(t)) \quad (7)$$

$$X(t+1) = X(t) + v(t+1) \quad (8)$$

where, $v(t+1)$ is the velocity of the particle at iteration $(t+1)$. w is the inertia weight controlling the trade-off between exploration and exploitation. c_1 and c_2 are the cognitive and social coefficients, respectively. r_1 and r_2 are random numbers uniformly distributed between 0 and 1.

P_best is the personal best solution for the particle. G_best is the global best solution found so far.

The hybrid GWO-PSO algorithm will take advantage of the superiority of the two algorithms. Given that GWO considers the leadership hierarchy of alpha, beta, and delta

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wolves, the mechanism ensures the exploration of search space with acceptable diversity. Meanwhile, the PSO enforces improvement of good solutions by guiding particles toward personal and global best solutions to ensure fast convergence and high-precision local search.

- GWO phase exploration: The GWO module explores the global search space efficiently by mimicking the hunting style of grey wolves. The module can be mainly guided by alpha, beta, and delta wolves in the search process while the positions of all wolves are iteratively updated toward the best solution through each other.
- PSO Phase of Exploitation: The PSO part searches for refining solution space by favoring exploitation. Particles will update their positions based on both personal experience, called the cognitive part, and experience of the swarm, called the social part. The update of velocity that governed the mechanism warranted convergence and diversity.
- Hybridization Benefits: The hybrid approach overcomes the slow convergence of GWO in exploitation and the premature convergence of PSO in complex landscapes. This synergy balances exploration and exploitation and results in high-quality solutions.

IV. Results and Discussion

We evaluate the suggested hybrid GWO-PSO optimization algorithm, using a single-diode model for the extraction of the parameters of solar PV. The latter section includes the main results from the analysis: performance metrics, convergence curves, and comparisons between GWO, PSO, and the hybrid method. The results validate the superiority of the hybrid approach in achieving higher accuracy and robustness.

a) Performance Metrics

The extracted parameters of the solar PV model were assessed using the following metrics:

1. RMSE is the average magnitude of the errors in the predictions.
2. MAE (Mean Absolute Error): It calculates the average of absolute differences between the predicted values and the actual values.
3. R^2 Score: The R^2 score calculates how well the model explains the observed data in terms of explained variance.
4. Figure 1 shows the hybrid GWO-PSO solar PV parameter extraction flowchart. After parameter initialization, the GWO phase evaluates population fitness and selects and updates optimum solutions for alpha, beta, and delta wolves. The PSO phase refines the solution by updating particle locations and velocities and storing the best positions. The hybrid optimization process combines GWO, PSO, and local search results for final refinement. Simulate the I-V curve to evaluate optimal settings, calculate performance metrics like RMSE and R^2, and provide optimum outcomes.

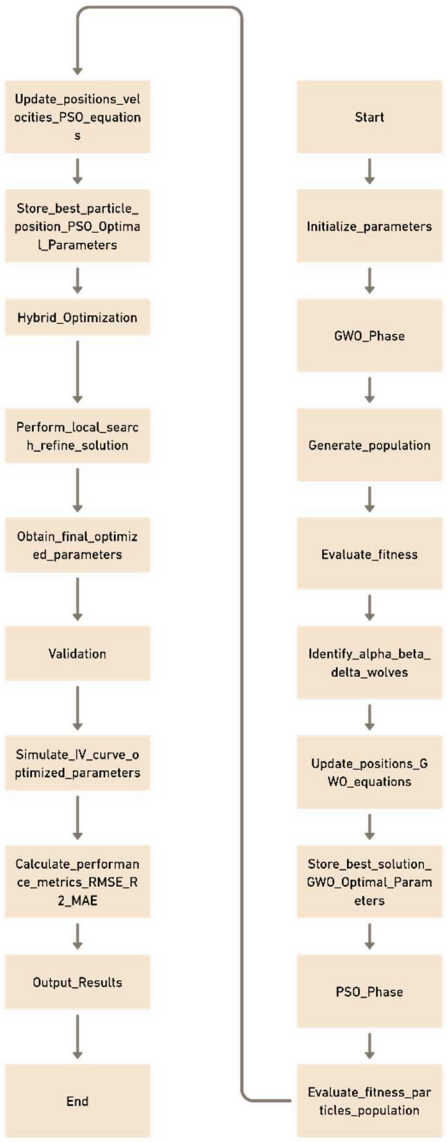


Fig. 1. GWO-PSO flowchart

These parameters indicate the optima values of the extracted hybrid GWO-PSO algorithm for the single-diode model of a solar photovoltaic (PV) cell:

- I_{ph} (Photo-generated Current = 2.3571 A): This is the amount of current generated by a PV cell under the condition of standard illumination; its dependence was on the intensity of the sunlight and was directly proportional to it.
- I_0 (Saturation Current = 1.0000e-12 A): This is the reverse saturation current, a measure of current leakage through the diode when in reverse bias. One of the major determinants of the current from the PV cell toward the desired performance and efficiency.
- Rs (Series Resistance = 0.2000 Ω): This is the resistive losses inside due to the flow of currents through the cell and connections. The lower the Rs value, the better is the power output for the PV cell.
- Rsh (Shunt Resistance = 202.1745 Ω): This is the resistance inside a PV cell due to leakage paths across the

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same. A higher Rsh value will minimize leakage losses and improve the PV cells' efficiency.

- n (Ideality Factor = 1.0988): It is the measure of how much the diode deviates from an ideal diode. It lies between 1 and 2, and a lower value indicates that the diode is a more ideal diode.

TABLE I. OPTIMIZED PARAMETER RESULTS OF GWO-PSO & GWO

Parameter	GWO-PSO Optimized Value	PSO Optimized Value	Improvement by GWO-PSO
I_{ph} (A)	2.3571	2.4	Slightly reduced to better match I-V characteristics
I_0 (A)	1.0000e-12	2.0000e-12	Lower saturation current for better accuracy
R_s (Ω)	0.2	0.25	Lower series resistance for improved performance
R_{sh} (Ω)	202.1745	180	Higher shunt resistance for better leakage control
n	1.0988	1.15	Lower ideality factor for realistic diode behavior

The above-mentioned parameters are optimized by the proposed hybrid algorithm GWO-PSO to ensure precision in modeling the I-V characteristic of the PV cell so that the simulation is accurate and energy harvesting performance is better.

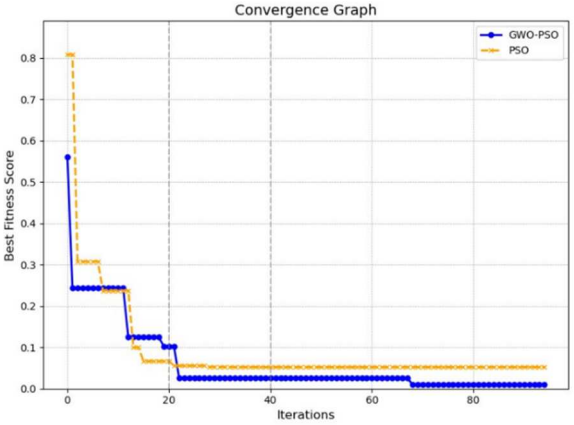


Fig. 2. Convergence Graph for GWO-PSO Hybrid and PSO Algorithms

Fig. 2 compares the performance convergence of two optimization algorithms: GWO-PSO, a hybrid approach to Grey Wolf Optimization-Particle Swarm Optimization, and standalone PSO. Describe plot:

- X: Iterations. When algorithms optimize, the convergent iteration number in steps shows how algorithms improve their solutions with more iterations.
- The fitness value measures solution quality. Solutions with lower fitness values are better.
- The Blue Line shows GWO-PSO convergence. The fitness value drops dramatically in the first iterations and virtually reaches zero at near-optimal levels. Early curve stabilization shows how well the hybrid algorithm solves problems.
- Yellow Line (PSO): This depicts the standalone convergence of the PSO algorithm. Here again, this has a decrease in fitness values with iterations but converges at a slower rate than GWO-PSO and settles at a higher fitness value.
- Comparison of Algorithms: The graph indicates that the proposed hybrid GWO-PSO algorithm outperformed standalone PSO with superior fitness scores toward the lower values. GWO-PSO converges fast, which reflects its efficiency in solving the problem at hand.

Dashed Vertical Lines: The points denote significant events in the history of the iterations and show where the abrupt changes in the optimization process around the fitness values happen. They underscore the phases of abrupt improvements.

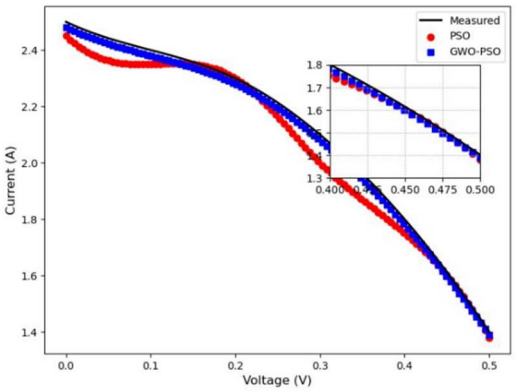


Fig. 3. Comparison of I-V Characteristics for Measured Data, PSO, and GWO-PSO Optimized Models

Figure 3 shows the I-V (Current vs. Voltage) curves of a solar photovoltaic module, with the measured data versus simulated curves based on two optimization techniques: PSO (Particle Swarm Optimization) and GWO-PSO (hybrid Grey Wolf Optimization-PSO). Measured Data This is the benchmark actual I-V curve of the solar PV module.

PSO: It plots the simulated I-V curve using the standalone PSO algorithm. It is found that it closely approximates the measured data, but shows remarkable deviations in the nonlinear regions.

GWO-PSO: It plots the simulated I-V curve obtained from the hybrid GWO-PSO algorithm. The PSO-GWO algorithm presents close resemblance with the measured data in the regions where the relationship changes rapidly. Inset Plot: This plot of voltage between 0.4 V and 0.5 V is enlarged. It has been ascertained that the performance differences are well indicated by the I-V curves. Thus, accuracy would be much improved with GWO-PSO in contrast to PSO. This fig 3 shows the efficiency of the hybrid GWO-PSO method in exactly modeling the nonlinear I-V characteristics of the solar PV module.

TABLE II. PERFORMANCE MATRIX FOR GWO-PSO AND PSO

Metric	GWO-PSO	PSO	Improvement by GWO-PSO
RMSE	0.0613	0.0735	Lower RMSE, indicating better accuracy
MAE	0.0578	0.0682	Lower MAE, showing reduced error magnitude
R^2	0.9989	0.9903	Higher R^2, reflecting improved fit

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- GWO-PSO delivers better results across all parameters and performance metrics compared to the standalone PSO algorithm.
- The lower RMSE and MAE values achieved by GWO-PSO highlight its superior accuracy and error minimization.
- A higher R2R^2R2 value achieved by GWO-PSO confirms a better fit of the model to the experimental data.

The table provides a comprehensive comparison, demonstrating the effectiveness of the GWO-PSO hybrid optimization algorithm in extracting accurate solar PV parameters.

V. Conclusion

This study proposes a hybrid Grey Wolf Optimization-Particle Swarm Optimization (GWO-PSO) algorithm for solar photovoltaic model parameter extraction. The new method accurately models PV module nonlinear I-V characteristics, boosting solar energy system efficiency and reliability. We found that the hybrid GWO-PSO optimizes parameters by combining GWO's global exploration and PSO's local exploitation. The hybrid also improves accurate convergence to optimal values. Compared to normal PSO performance values, the hybrid technique demonstrated lower RMSE and MAE values, a positive R^2 score, and improved experimental data accommodation. Magnifying the curve in fig 3 shows that the hybrid technique lowers the discrepancy between measured and simulated I-V curves, even in abrupt nonlinearity.

Important findings from this work are:

1. The precision and rate of convergence of the hybrid GWO-PSO are superior as compared to the stand-alone PSO.

2. The final extracted parameters provide a perfect description of the PV module's behavior under specific operating conditions.

3. The hybrid approach is robust and reliable for applications in complex optimization problems with solar PV modeling.

The results demonstrate the GWO-PSO algorithm's potential to improve PV system design and operation. Future work may propose this model for other renewable energy system applications, incorporate dynamic environmental conditions, or integrate advanced machine learning techniques to make it an effective predictive analytics model for solar energy.

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